

# Predicting Income Level (>\$50K): Executive Report

## Business Goal

Predict whether an individual earns more than \$50,000/year from census-style demographic and employment attributes, to support a targeted outreach program. Because false positives directly waste outreach spend, the project is precision-first: the model must be confident before flagging someone as a high earner, even at the cost of missing some true positives.

## Data Used

UCI Adult / Census Income dataset (48,842 records). Features: age, workclass, education level, marital status, occupation, relationship, race, sex, capital gain/loss, hours worked per week, and native country, plus engineered features (education-hours interaction, age/hours buckets, capital-gain/loss indicators, net capital). Stratified 70/12.5/17.5 train/dev/test split (random\_state=42) to preserve the ~24% positive class rate across all splits.

## Final Model & Deployment Artifact

Histogram Gradient Boosting Classifier (scikit-learn), tuned via randomized search (learning\_rate=0.0129, max\_depth=9, max\_iter=108, l2\_regularization=1.0256), wrapped in an isotonic CalibratedClassifierCV so output probabilities are well-calibrated rather than raw scores. Three class-imbalance strategies (class\_weight='balanced', random oversampling, SMOTE) were each tested inside leakage-safe pipelines via 5-fold CV; none improved discriminative power (Average Precision 0.79-0.80 across all variants), so the **unmodified baseline pipeline was retained** for simplicity, training speed, and dependency footprint. The model, preprocessing pipeline, and calibration layer are serialized (joblib) and served via a standalone inference script (inference.py) with input validation and unit tests.

## Key Hold-Out Metrics

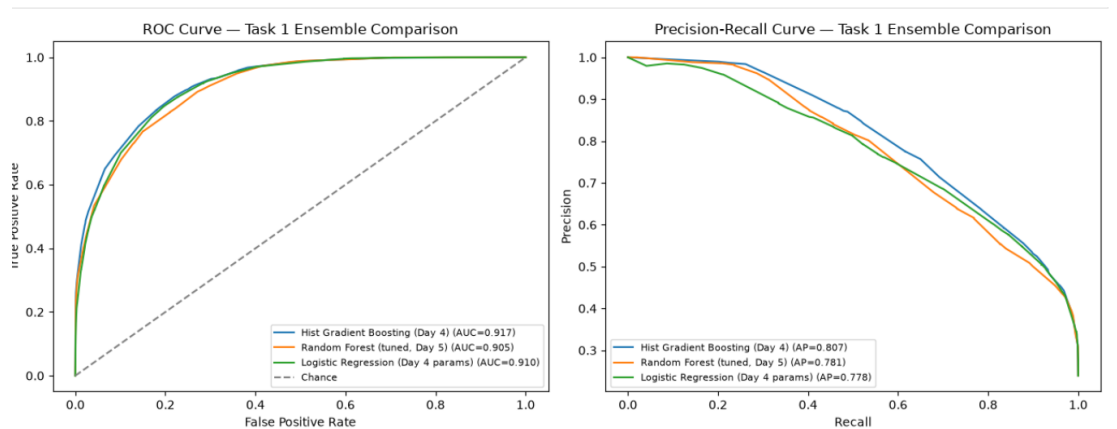
| Metric             | Value  | Day 4 Report (cross-check) |
|--------------------|--------|----------------------------|
| Precision          | 0.8417 | 0.8425                     |
| Recall             | 0.5231 | 0.5240                     |
| ROC AUC            | 0.9173 | 0.9173                     |
| Decision threshold | 0.57   | 0.57                       |

Threshold 0.57 (not the naive 0.5 default) was deliberately tuned in Day 4 to favor precision, consistent with the project's cost framing.

## Ensemble Comparison (Task 1)

| Model                              | Precision | Recall | ROC AUC | PR-AUC | Inference (ms/row) |
|------------------------------------|-----------|--------|---------|--------|--------------------|
| Hist Gradient Boosting (final)     | 0.8440    | 0.5205 | 0.9173  | 0.7944 | 0.0361             |
| Random Forest (tuned)              | 0.8134    | 0.5073 | 0.9055  | 0.7616 | 0.0119             |
| Logistic Regression (Day 4 params) | 0.7927    | 0.5218 | 0.9102  | 0.7604 | 0.0052             |

HGB leads on precision (the project's primary metric) and PR-AUC. Logistic Regression is competitive on recall and by far the fastest at inference, but trails on precision. Random Forest underperforms both alternatives on nearly every metric. HGB is retained as the production model on the strength of its precision advantage.



ROC and precision-recall curves, hold-out test set. The PR curve shows the clearest separation (HGB pulling ahead of RF and Logistic Regression through the mid-recall range), and is the more informative view given the dataset's ~24% positive class rate.

## Interpretability Findings

Permutation importance (scored on Average Precision) and SHAP mean  $|\text{value}|$  independently converge on the same top 8 features: capital-gain, marital-status (married-civ-spouse), education-hours interaction, capital-loss, age, education-num, occupation (Other-service), and workclass (Self-emp-not-inc). Marital status produces the cleanest split in the SHAP summary; a small subset of individuals with large capital gains receive a decisive push toward >50K; capital-loss counterintuitively pushes predictions upward, functioning as a wealth proxy rather than a hardship signal in this dataset. Local case studies (one true positive, one false positive, one false negative) confirmed the model leans heavily on the combination of marital status, education level, and hours worked, sometimes over-weighting credentials relative to actual reduced-hours income (false positive: a semi-retired professional) and under-weighting stable public-sector roles without formal degrees (false negative).

## Fairness Observations

By sex: Precision is higher for women than men (90.6% vs. 83.6%), but recall is 15 points lower (39.5% vs. 54.4%) — the model is disproportionately conservative about predicting >50K for women, plausibly reflecting the strong correlation between the model's top features (marital-status, relationship=Husband) and sex. By race: Asian-Pac-Islander individuals see the lowest precision of any group (73.1% vs. 80-86% elsewhere) despite a base rate nearly identical to White individuals, indicating disproportionate false positives for that group. Amer-Indian-Eskimo and Other groups show lower recall but with small sample sizes ( $n=104$ ,  $n=69$ ) that warrant caution rather than firm conclusions.

## Proposed mitigations:

- (1) evaluate group-specific threshold calibration, particularly for Asian-Pac-Islander precision;
- (2) audit marital-status/relationship features as sex proxies, testing a gender-neutral re-encoding; (
- (3) prioritize additional data collection for small-n race groups;
- (4) report precision/recall by group in ongoing monitoring, not just the aggregate.

## Recommended Next Steps

- A/B test plan: Route a held-out slice of new outreach candidates through the model vs. the existing (non-model) targeting process for 4-6 weeks; compare precision of flagged individuals and cost-per-successful-outreach between arms before full rollout.
- Monitoring: Track data drift (PSI/KS on top features, weekly), prediction-score distribution (daily), and precision/recall by group (rolling window, once labels arrive); alert if precision drops below 0.75 or group disparities widen beyond current baselines.
- Retraining cadence: Scheduled every 6 months, or immediately upon a triggered alert; re-run the imbalance comparison and fairness check at every retraining cycle.
- Fairness remediation: Pilot the marital-status/relationship re-encoding proposed above and re-evaluate the sex recall gap before the next production release.